WorkRadar PRD

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WorkRadar gives an engineering manager one screen that answers four questions in under a minute: what to work on today, what happened, what needs attention now, and who is blocked, without opening Slack, Jira, GitHub or email.

Status: built and live. App: work-radar-theta.vercel.app. Code: github.com/BikramSailorBoy/WorkRadar. Against the acceptance criteria, 11 of 11 planted signals surface in the correct panel, with 0 noise leaks. On the live deploy, the AI brief takes 3.8 seconds.

Problem and goal

A manager's day starts with rebuilding one picture from four or more tools. The signal is scattered: a blocker is mentioned in Slack but never flagged in Jira, an escalation arrives by email, and a pull request waits on a reviewer who is on leave. Each tool is accurate on its own, but none of them answers the manager's actual question: where is my attention needed?

Goal. From one screen, in under 60 seconds, a manager can answer the four questions and act on the top items.

Non-goal. WorkRadar is not a productivity scorer. It ranks work and blockers, never people. That line is deliberate: a tool that ranks individuals erodes the trust a manager needs to unblock their team.

User

Primary persona: Marty McFly, Engineering Manager at Darb Pay (fictional), headquartered in Cologne on the Rhine. His time-travelling days are behind him. His evenings now belong to Jennifer, so his mornings have to count. He runs 8 engineers in two squads, Payments and Onboarding, spread across Riyadh, Cairo and Bengaluru. He works Monday to Friday on Cologne time, while his Riyadh and Cairo engineers work Sunday to Thursday. When he logs in at 09:30, Bengaluru has been at it for four hours, and the Riyadh team has already had a full Sunday he never saw.

Job to be done. When I start my day, I want to know where my attention is needed and who I can unblock, so that standup is about decisions, not status.

Today, without WorkRadar	With WorkRadar
Scans Slack channels and DMs for anything urgent	Reads a 60-second brief where every line links to its source
Filters Jira for blocked and overdue tickets	Sees blocked people and the chain behind each block
Checks GitHub for PRs waiting on him or stuck in review	Gets his own pending decisions and reviews in Top work
Searches email for customer or vendor escalations	Sees escalations ranked by impact in Needs attention
Pieces together what changed overnight	Drags the replay bar to watch the day so far

Secondary users (later): a tech lead covering for the manager, and a director scanning several teams.

The four questions

Each question has a written rule, so two people looking at the same data get the same answer. Rules decide what qualifies; AI reads the unstructured text the rules cannot.

Question	What counts	Rule (deterministic)	What AI adds	Shown as
Top work for the day	Up to 5 open items that need progress today, including the manager's own reviews and decisions	Highest priority score; blocked items move to the Blocked panel	A one-line reason per item, naming the factors behind its score	Ranked list with score chip
What happened today	Meaningful changes since the team's working day began (08:00 Riyadh, 07:00 Cologne), or since the last visit	Merges, deploys, items moved to Done, new incidents, new escalations; CI, bot and reaction events filtered out	A 3 to 5 sentence brief, every sentence citing its sources	Morning brief and replay timeline

Question	What counts	Rule (deterministic)	What AI adds	Shown as
Needs attention now	Items where waiting costs money or trust today	Score of 75 or more, plus one of: P0/P1 incident, customer escalation, due within 24 hours, decision requested from the manager	Spots escalation tone, decision requests and priority conflicts in Slack and email	Inner ring of the radar; first in triage queue
Who is blocked	A person who cannot progress an item because of someone or something else	Jira status Blocked; PR waiting on review for over 24 working hours; depends on an item that is itself blocked	Finds blockers mentioned only in chat or email: who or what blocks, and since when	Person cards with the blocker chain behind each

Working hours exclude each owner's weekend: Friday and Saturday in Riyadh and Cairo, Saturday and Sunday in Bengaluru and Cologne. All times show in the manager's time zone.

Experience

One screen, read top to bottom: the brief tells the story, the radar shows its shape, the four panels carry the detail, and triage turns it into action.

Visual direction: a radar command console. The screen is styled as a holographic combat information center. The radar sits in the centre, and six compact widgets surround it: Needs attention, Who is blocked and Morning brief on the left; Top work, What happened and Ask on the right. Each widget shows a number and one line, and expands on click to show the full detail, so the first screen stays readable. Every number on screen comes from the data; there are no decorative readouts.

Noise as sea clutter. Real radars show faint speckle from waves, which operators suppress. WorkRadar draws the 121 bot, CI, reaction and chit-chat events as dim clutter, behind a Clutter suppression toggle that is on by default. Turning it off shows how much the product filters out for the manager.

Morning brief and Ask. At the top, three to five sentences summarise the day. Each sentence ends with source chips, such as PAY-412 or PR #318, and clicking a chip highlights that item everywhere. Beside it, an Ask box takes free-text questions ("what's slipping before the freeze?") and answers only from today's data, with citations. If the answer is not in the data, it says so.

The radar (hero). Every open item is a blip.

- **Sector:** the source tool. Jira, GitHub, Slack and email each own a quadrant.
- **Distance from centre:** urgency. The closer the blip, the higher the priority score. The inner ring is "Act now" (75+), the middle ring "Today" (50 to 74), the outer ring "Watch".
- **Shape:** blocked items are hollow with a dashed outline, so status never relies on colour alone.
- **Motion:** items that changed in the last hour pulse once, then settle.
- **Hover and click:** hovering shows title, owner, score and the AI reason; clicking opens a drawer with the item's event trail across tools.

The four panels. Top work, What happened, Needs attention and Who is blocked sit beside the radar. They hold the same items as the radar, so everything the radar shows is also reachable by keyboard and screen reader.

Day replay. A scrubber runs along the bottom from the start of the work week to now, with ticks for each meaningful event, coloured by source. Drag it, and the radar and panels rebuild to that moment: watch the incident appear at 07:40 and the KYC release land at 08:15, Cologne time. A play button runs the day at one hour per second.

Triage mode. Press T, and the attention and blocked items come up one at a time.

- **A** acknowledges.
- **D** delegates to a team member.
- **S** snoozes until tomorrow.
- **N** nudges: AI drafts a short message to the owner, which the manager edits and copies.

A counter shows progress ("4 of 7 triaged"). Decisions persist in the browser for the session. Write-back to Jira or Slack is out of scope for the MVP.

Live or demo indicator. A small badge always shows whether AI outputs are live or precomputed.

Prioritization and AI design

The design principle: rules rank, AI reads and explains. The model never invents a score. The score is computed from data, so it is reproducible and auditable, and the AI's job is to understand the unstructured text that rules cannot read.

Pipeline

1. **Normalise.** Events from the four sources become one schema: person, work item, event.
2. **Extract (Claude Haiku 5.5).** Each Slack thread and email is classified for:

- a blocker: who or what is blocking, and since when
- an escalation
- a decision requested from the manager
- the work items it mentions

Output is strict JSON checked against a schema. Anything invalid is dropped and logged.

3. **Score (code).** Each item gets a priority score from 0 to 100, from the factors below.
4. **Explain (Claude Haiku 5.5).** Each top item gets a one-line reason, built from its highest-scoring factors.
5. **Brief and Ask (Claude Sonnet 5.5).** These are grounded only on the derived state. Every sentence must cite item IDs. A post-check removes any sentence without a valid citation.

Priority score

Factor	Max points	How it is computed
Urgency	30	Due within 24h: 30. Within 3 days or before the release freeze: 20. This sprint: 10. Open P0 incident: 30
Impact	25	P0 25, P1 18, P2 10, P3 4; +5 if customer-facing, capped at 25
Blocking fan-out	20	7 per item or person waiting on it, capped at 20
Staleness	15	No movement for over 72 working hours: 15. Over 24: 8
Escalation signal	10	AI-flagged escalation, an executive mention, or a decision requested from the manager

Guardrails

- **Citations required and validated.** A claim without a source is not shown.
- **No person-level ranking.** People appear only as owners of work or as blocked.
- **Bounded prompts.** Prompts say "answer only from the data provided; otherwise say it is not in today's data".
- **Structured outputs.** Extraction returns JSON validated before use.

Fallback. If there is no API key, or a call fails or takes longer than 20 seconds, the app serves precomputed outputs. These are generated by the same prompts on the same data and labelled "Demo mode", so the app works for anyone who opens it.

Model choice. Haiku handles the high-volume, short, structured tasks; Sonnet handles synthesis, where quality matters most. The cost per brief will be measured and reported in the AI Workflow document.

Data

The mock data is event-sourced. Every change is a timestamped event, and the dashboard at any moment is derived by replaying events up to that moment. That one decision makes Day replay cheap and correct, and it mirrors how real webhooks from Slack, Jira and GitHub arrive.

What's in it. 15 people, 45 work items and 246 events. The items are 18 Jira tickets including one incident, 9 pull requests, 14 Slack threads and 4 emails. They run from Sunday 11 Oct 08:00 to a fixed demo clock of Tuesday 13 Oct 10:30 Riyadh time (09:30 in Cologne), so every demo shows the same day. All names, companies and banks are fictional.

Planted signals. Twelve situations are written into the data on purpose, and they serve as ground truth. An automated check counts how many of them WorkRadar surfaces in the right panel (recall), and how much noise leaks into Top work or Needs attention (precision). Roughly 45% of events are noise.

#	Signal	Where the clue hides	Should appear in
S1	P0 incident: card payments failing for Northstar Bank	Incident ticket and #incidents thread	Needs attention
S2	PR #318 waiting about 30h on a reviewer who is out of office	GitHub review request and an OOO note in Slack	Who is blocked
S3	PAY-412 blocked on KYC vendor keys; vendor's question unanswered for 2 days	Jira status and an unanswered email	Who is blocked
S4	Arjun waiting on staging DB credentials from infra	Slack only; the Jira ticket still says In Progress	Who is blocked
S5	Palmway Retail CFO escalates settlement delays	Email, linked to PAY-398	Needs attention
S6	ONB-77 is P3 in Jira, but the CEO needs it for Thursday's investor demo	Slack message from the CEO	Top work

#	Signal	Where the clue hides	Should appear in
S7	ONB-70 In Progress for about 4 days with no activity	Absence of events	Needs attention
S8	KYC v2 onboarding flow merged and deployed	GitHub merge and deploy, Jira to Done	What happened
S9	Release freeze starts Thu 15 Oct 12:00	#eng-announcements	Top work
S10	Blocker chain: ONB-81 waits on PAY-405, which waits on infra	Jira links and a Slack thread	Who is blocked
S11	CI passes, bot posts, reactions, chit-chat, P3 grooming	Everywhere	Nowhere (noise)
S12	Sara needs Marty's scope-vs-date decision; PR #322 waits 20h on his review	Slack DM and GitHub	Needs attention

Options considered

Three visual concepts were weighed against a plain list, judged on four tests: does it answer the four questions, can it be read at a glance, is it precise, and can it be built well in one day?

Concept	How it answers the four questions	Strength	Weakness	Verdict
Radar with day replay and triage	Distance shows urgency, sector shows source, hollow blips show blocked items, replay shows what happened, triage closes the loop	Shape of the day at a glance; replay makes "what happened" visible; triage turns insight into action	A radar can become decoration if positions are not tied to a real score	**Chosen.** Every blip position comes from the priority score

Concept	How it answers the four questions	Strength	Weakness	Verdict
Air traffic control	Landing queue = top work, arrivals board = what happened, low on fuel = needs attention, holding pattern = blocked	The strongest metaphor for waiting: "circling for three days" lands instantly	Heavier to build; dense on small screens; the metaphor needs a moment to learn	**Strong alternative.** The holding-pattern language informs the Blocked panel copy
Team weather forecast	Conditions per squad (clear, cloudy, storm warning) and a pressure map of load	Instant emotional read; works well on mobile	Imprecise for "top work"; weather per person reads like surveillance	**Rejected** for the MVP; may suit a director's multi-team view later
Plain prioritised lists	Four ranked lists	Fastest to build; most precise	No differentiation; hard to see the shape of the day	**Baseline.** Kept inside the chosen design as the four panels, which also serve as the accessible path

Scope

The MVP proves the core loop on mock data: see, understand, act. Live integrations wait until that loop works.

In the MVP (this submission)

- The four panels, the radar, day replay and triage mode, on mock data
- AI extraction, per-item reasons, the morning brief and Ask, all calling Claude live
- Demo mode with precomputed AI outputs when no key is set
- A deployed public URL, plus a repo that runs locally
- A ground-truth check against the 12 planted signals

Later

- OAuth connectors for Slack, Jira, GitHub and Gmail or Outlook, with webhooks
- Write-back from triage: assign a reviewer, comment, update status
- The brief pushed to Slack or WhatsApp at the start of the day
- Multi-team director view, where the team weather concept may return
- Learning from triage: snoozed items rank lower next time
- Per-team working hours, holidays and sprint calendars

Success metrics

The prototype is judged against its own ground truth. A pilot would then be judged on time saved and blockers cleared.

Acceptance criteria for the prototype

Criterion	Target
Planted signals surfaced in the correct panel (recall)	11 of the 11 non-noise signals, minimum 10
Noise items in Top work or Needs attention	0
Brief sentences carrying a valid source citation	100%
Hidden blocker S4 found from Slack text alone	Yes
Works with no API key	Yes, in demo mode
Dashboard first load in demo mode	Under 2 seconds
Live morning brief generated	Under 8 seconds

Product metrics for a pilot

- Time to answer the four questions: under 60 seconds, measured in moderated sessions
- Tools opened before first standup: down from the current baseline, measured by self-report
- Median blocker age: falling week on week
- Triage completion: share of attention items acknowledged, delegated or nudged the same day
- AI reason quality: under 10% of reasons marked unhelpful

Baselines for the pilot metrics would be captured in week one; none are assumed here.

Architecture

A single Next.js app holds the UI, the derivation logic and the AI routes. The API key stays on the server.

1. **Data layer.** People, items and events load as JSON from the repo.
2. **Derive (code).** Given a time t, it replays events up to t and returns the board state. Replay calls the same function with a different t.
3. **Score (code).** It applies the priority formula and assigns rings and panels.
4. **AI routes (server).**
   - `/api/extract`: Haiku signal extraction
   - `/api/reasons`: Haiku one-line explanations
   - `/api/brief`: Sonnet brief
   - `/api/ask`: Sonnet Ask

   Each validates its output and falls back to the precomputed cache.
5. **UI.** React, TypeScript and Tailwind CSS. The radar is hand-built SVG, and the panels, drawer, triage and replay are client components.
6. **Hosting.** Vercel, with `ANTHROPIC_API_KEY` set as a server-side environment variable.

In production the data layer would be replaced by connector webhooks writing to an event store. The derive, score and AI layers stay the same.

Risks and open questions

The biggest risk is trust. If the AI states one thing that is not true, the manager stops relying on the screen.

Risk	Mitigation
AI states something that is not in the data	Rules own the ranking; every AI sentence carries a validated citation or is dropped
Feels like surveillance of the team	No person-level scores; people appear only as owners of work or as blocked, and the copy is framed around unblocking
API cost, latency or outage	Haiku for high-volume calls; cached results; demo mode as an automatic fallback
Radar becomes decoration	Every blip position is derived from the score; the four panels carry the full answer

Risk	Mitigation
Mock data is too convenient	45% noise, a hidden blocker and a priority conflict are built in; real connectors would add OAuth, rate limits and messier text

Open questions

- For a team spread across three time zones, when does "today" start: the manager's morning or each person's?
- Will managers send AI-drafted nudges as written, or do they want to edit every one?
- Should snoozed items come back on a schedule, or only when something about them changes?